



```
In [1]: import numpy as np
        from tensorflow.keras.datasets import imdb
        from tensorflow.keras.layers import Dense, Input
        from tensorflow.keras.models import Sequential
```

```
In [2]: (train_data, train_label), (test_data, test_label) = imdb.load_data(num_words = 10000)

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 ————— 0s 0us/step
```

```
In [3]: def vectorize_sequence(sequence, dimension=10000):
        results = np.zeros((len(sequence), dimension))
        for i, seq in enumerate(sequence):
            results[i, seq] = 1
        return results
```

```
In [4]: x_train = vectorize_sequence(train_data)
        x_test = vectorize_sequence(test_data)
        y_train = np.asarray(train_label).astype('float32')
        y_test = np.asarray(test_label).astype('float32')
```

```
In [5]: model = Sequential()
        model.add(Input(shape=(10000,)))
        model.add(Dense(64, activation='relu'))
        model.add(Dense(32, activation='relu'))
        model.add(Dense(1, activation='sigmoid'))
```

```
In [6]: model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

```
In [7]: history = model.fit(x_train, y_train, epochs=20, batch_size=512, validation_split=0.1)
```

Epoch 1/20
35/35  4s 80ms/step - accuracy: 0.8194 - loss: 0.4454 - val_accuracy: 0.8845 - val_loss: 0.2907
Epoch 2/20
35/35  2s 60ms/step - accuracy: 0.9188 - loss: 0.2115 - val_accuracy: 0.8921 - val_loss: 0.2761
Epoch 3/20
35/35  2s 58ms/step - accuracy: 0.9557 - loss: 0.1352 - val_accuracy: 0.8871 - val_loss: 0.2974
Epoch 4/20
35/35  2s 58ms/step - accuracy: 0.9741 - loss: 0.0912 - val_accuracy: 0.8805 - val_loss: 0.3350
Epoch 5/20
35/35  3s 61ms/step - accuracy: 0.9871 - loss: 0.0566 - val_accuracy: 0.8783 - val_loss: 0.3801
Epoch 6/20
35/35  3s 77ms/step - accuracy: 0.9949 - loss: 0.0321 - val_accuracy: 0.8755 - val_loss: 0.4330
Epoch 7/20
35/35  4s 51ms/step - accuracy: 0.9984 - loss: 0.0171 - val_accuracy: 0.8727 - val_loss: 0.4808
Epoch 8/20
35/35  3s 51ms/step - accuracy: 0.9996 - loss: 0.0101 - val_accuracy: 0.8728 - val_loss: 0.5212
Epoch 9/20
35/35  3s 51ms/step - accuracy: 0.9998 - loss: 0.0058 - val_accuracy: 0.8720 - val_loss: 0.5563
Epoch 10/20
35/35  3s 73ms/step - accuracy: 1.0000 - loss: 0.0036 - val_accuracy: 0.8719 - val_loss: 0.5884
Epoch 11/20
35/35  2s 59ms/step - accuracy: 1.0000 - loss: 0.0025 - val_accuracy: 0.8728 - val_loss: 0.6129
Epoch 12/20
35/35  2s 51ms/step - accuracy: 1.0000 - loss: 0.0019 - val_accuracy: 0.8719 - val_loss: 0.6354
Epoch 13/20
35/35  3s 50ms/step - accuracy: 1.0000 - loss: 0.0014 - val_accuracy: 0.8719 - val_loss: 0.6537
Epoch 14/20
35/35  2s 51ms/step - accuracy: 1.0000 - loss: 0.0012 - val_accuracy: 0.8719 - val_loss: 0.6733
Epoch 15/20
35/35  2s 60ms/step - accuracy: 1.0000 - loss: 9.4576e-04 - val_accuracy: 0.8717 - val_loss: 0.6892
Epoch 16/20
35/35  3s 73ms/step - accuracy: 1.0000 - loss: 7.8349e-04 - val_accuracy: 0.8720 - val_loss: 0.7037
Epoch 17/20
35/35  2s 49ms/step - accuracy: 1.0000 - loss: 6.5823e-04 - val_accuracy: 0.8712 - val_loss: 0.7188
Epoch 18/20
35/35  3s 59ms/step - accuracy: 1.0000 - loss: 5.6044e-04 - val_accuracy: 0.8712 - val_loss: 0.7330

Epoch 19/20

35/35 ————— **2s** 51ms/step - accuracy: 1.0000 - loss: 4.8067e-04 -
val_accuracy: 0.8715 - val_loss: 0.7457

Epoch 20/20

35/35 ————— **3s** 66ms/step - accuracy: 1.0000 - loss: 4.1462e-04 -
val_accuracy: 0.8716 - val_loss: 0.7579

```
In [8]: loss, accuracy = model.evaluate(x_test, y_test)
        print("Test Accuracy:", accuracy)
```

782/782 ————— **2s** 3ms/step - accuracy: 0.8585 - loss: 0.8205
Test Accuracy: 0.858519971370697

```
In [9]: model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	640,064
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 1)	33

Total params: 1,926,533 (7.35 MB)

Trainable params: 642,177 (2.45 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 1,284,356 (4.90 MB)